

Math 199, Spring 2022
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Participation Assignment 2 - Adapting trigonometric substitution

Estimated Time: Less than 1 hour (for first two pages).

Goals: The main goals of today's class will be to learn to adapt the method of trigonometric substitution to various contexts.

1) Recall the idea behind trigonometric substitution from last week for the integral: $\int \sqrt{1-x^2} dx$. Draw and label the relevant triangle in terms of both x and θ .

2) Highlight the trig identity at the core of your answer to the previous problem.

The main observation to make for the trig sub method, is that your answer to problem (2) is really an instance of the Pythagorean theorem

$$a^2 + b^2 = c^2$$

for a right triangle with legs a , b , and hypotenuse c , which is an adaptable starting point that can be applied in different ways.

3) Use this observation to make an appropriate substitution to compute the indefinite integral: $\int \sqrt{a^2 - x^2} dx$. (Start by drawing and labeling a triangle; then re-label the triangle in terms of trig functions and an angle θ).

4) Use the trig identity from problem (2) to derive the following trig identities:

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

5) Draw triangles that correspond to the identities in problem (4), and re-label the sides of the triangle with the intention of finding a good substitution for integrals that contain the expression $\sqrt{1+x^2}$.

6) Adapt problem (5) to find substitutions for integrals involving an expression of the form $\sqrt{a^2+x^2}$.

7) Which of the above substitutions look to you to be more promising, based on your knowledge of taking anti-derivatives? Why?

8) Use the ideas above to find a substitution for integrals involving expressions of the form $\sqrt{x^2-a^2}$.

If you've gotten to this point, that's great! If you have more time before the break, try the following problems.

The 231 worksheet on trig integrals from last week gave some particular problems and then mentioned that such a technique works in some general case. In the following problems, I want you to consider some of these general forms of integrals, and determine which method(s) to use to compute each one. In these problems, m and n represent arbitrary integers.

Important: you may need to split into multiple cases where one or both of m and n are even or odd!

9) $\int \sin^m x \cos^n x dx$

10) $\int \sec^m x \tan^n x dx$

11) $\int \csc^m x \cot^n x dx$